Data management plan

F01 Data management plan v2.0 – RDM

For requirements, explanation of the topics, definitions and abbreviations, see [SOP 001 Research Data Management – RDM – Amsterdam UMC](). The numbers in this DMP template refer to the numbers in the SOP 001 RDM.

The following study info provides relevant information regarding data management requirements:

Study number (ABR/METC)	NL84795.018.23
Study (acronym or short title)	The Netherlands ME/CFS Cohort and Biobank (NMCB)
Sponsor (Verrichter/initiator/coördinerende hoofdonderzoeker)	Amsterdam UMC, location AMC
Department	Department of Medical Psychology
Mono/Multicenter & (inter)national	Monocenter
(non)WMO	☒ WMO compliant ☒ not WMO compliant
Summary of the project	*The Netherlands ME/CFS Cohort and Biobank (NMCB) consortium, in partnership with patient organizations, is leading the way for the development of a national research infrastructure for Myalgic Encephalomyelitis and Chronic Fatigue Syndrome (ME/CFS). ME/CFS is a condition that disrupts patient's lives and causes great suffering with an estimated 40,000-150,000 patients in the Netherlands. The necessity for this initiative has been recognized by the Dutch Health Council (Gezondheidsraad). A national infrastructure for data and sample collection will increase the investment in biomedical research into ME/CFS. This need is amplified by the stigmatization of ME/CFS and its significant socio-economic impact. Adhering to ZonMw's "ME/CFS Research Agenda", the core activity of the NMCB is to establish a national patient cohort and biobank at the Amsterdam UMC while building a collaborative research infrastructure. This infrastructure will enable the validation of biomedical hypotheses for a profound mechanistic insight into ME/CFS and other Post- Acute Infectious Syndromes (PAIS), fostering rational diagnostics and effective treatments. Leveraging the collective expertise of all 7 Dutch University Medical Centers in the consortium, the NMCB prioritizes open access and practices FAIR data management, ensuring the data and biological materials are utilized to enhance patient well-being and societal integration. Recently this consortium was provided additional funding to extend its portfolio to include post-COVID.*

Co-ordinating PI (Hoofdonderzoeker)	Jos A. Bosch, PhD (j.a.bosch@amsterdamumc.nl)
Responsible for completion of the data management plan	*Shuxin Zhang (s.x.zhang@amsterdamumc.nl)*
Funding by	ZonMw (Netherlands)
Consulted data management expert	*Manaï, M (m.manai@amsterdamumc.nl)*
Data management plan version / date	2.0 /December 3rd, 2024
PI signature for approval of this DMP	

Phase 1: Study preparation	
Privacy and security safeguards	
1.1	*☐ The data set is anonymous and cannot be linked to any subject* ***☒ The data set is encoded; a meaningless unique code (Subject ID) is used, and subjects can be identified through a subject identification log*** *☐ The data set is directly identifiable* *Explanation:* *Data are encoded (by assigning participant identification (ID) numbers to all included participants) to ensure the security of privacy-sensitive information and/or to get in touch with subjects during data collection. The assigned participant ID numbers are subsequently used to merge data that are collected over multiple time points and from multiple sources. The participant ID number also allows future additions to the data.*
1.2	☒ Data are de-identified within the study database (see 1.1) ☐ Additional de-identification steps are taken, *specify: …* *The personal identifiable information (PII) will be stored in a separate location with controlled access, thereby not included in the study database:* - *File name: NMCB_Subject_Identification_Log.xlsx* - *File location: G:\divjk\mp_oz\NMCB\E. Patient information form (PIF) and informed consent form (ICF)\0. Subject ID Log* *De-identification is only allowed by authorized personnel who have GCP certificate.*
1.3	☒ A Data Protection Impact Assessment (DPIA) has been performed, *specify document:* - *File: 20240523_DPIA_ Resultaat Privacy Impact Assessment (DPIA)- beleid AmsterdamUMC (Versie 5) NMCB.doc* - *Location: G:\divjk\mp_oz\NMCB\P. Privacy* ☒ The data acquisition has been registered at the DPO, *specify registration number of the record of processing activities (Verwerkingsregister): 157135* *The Medical Ethical Review Committee (METc) for WMO study has been approved.* *In addition, two BIV classification files are filled in to assess IT systems unauthorized:* - *File 1: 20240507_BIV Classificatie Formulier - Dienst ICT - Amsterdam UMC (Versie 4) NMCB_CommunicatieDatabase.xlsx* - *File 2: 20240507_BIV Classificatie Formulier - Dienst ICT - Amsterdam UMC (Versie 4) NMCB_InhoudDatabase.xlsx*
1.4	The study has been (pre)registered or a concept/design paper has been published, *specify registration number(s): In progsress.*
1.5	☒ An informed consent procedure has been set up that describes the data set, time span of data retention, information on sharing data or making data available for future research. *The informed consent will be filled in before any study procedures are initiated, on paper in twofold. The content of informed consent is derived from the CCMO and Amsterdam UMC biobank template files: G:\divjk\mp_oz\NMCB\ E. Patient information form (PIF) and informed consent form (ICF)* *The wet-ink signed ICFs will be stored in the ISSF behind lock and key.*
1.6	☒ A central location for all digital study documents and (references) to data exists; *specify:* *A central SharePoint server located at the Amsterdam UMC (A-UMC) and a departmental drive will store some digital study.* - *https://amsterdamumc.sharepoint.com/sites/CoreTeam* - *https://amsterdamumc.sharepoint.com/sites/NMCB* - *G:\divjk\mp_oz\NMCB* ☒ A central location for all hard copy study documents exists in *Rosalyn building Paalbergweg 2-4, 1105 AG Amsterdam, the Netherlands.*
Data acquisition	
General	
1.7	Describe the data acquisition per type of data ☒ Reuse of existing data, e.g., patient characteristics from EPIC, or data from other research; *specify:* - *Physician records such as medical history, etc.* - *Data from previous research, such as the NMCB screening study, Lyme cohort, MS cohort, long-COVID cohorts, post-Q-fever cohorts: when given explicit permission in the ICF by study subjects*

	☒ Use of measured data: - *Blood laboratory values* - *MRI* - *ECG* - *Heart rate* - *Blood pressure* - *Weight* - *Height* - *Bioimpedance* - *Handgrip strength* - *Pulse oximetry* - *Cognitive functioning* ☒ Data collection: - *Questionnaire data: DSQ-SF, DSQ-2, CAGE-AID, PHQ-2, COMPASS-31, KPS, RAND36, PSQI, ESS, painDETECT.* - *Biobank data such as the biomaterial type, size, and created date.*
1.8	*Describe the terminology standards, classifications or existing data definitions that have been applied in the data set:* *The following terminology standards, in order of priority, will be applied, though they could serve for different aspects of healthcare:* - *SNOMED CT: for general clinical concepts* - *LOINC: for lab data such as measurements* - *ATC: for medication* - *ICD-10 & Mondo: for disease name* - *HGVS: for genetic data*
1.9	☒ All acquired data, either reused, measured, or manually collected, are described in a data dictionary or code book *A data dictionary will be built to describe data elements (or variables), their description, data type, permitted value, terminology, and constraints for validation:* - *File name: "NMCB Codebook version 2.xlsx"* - *Location: G:\divjk\mp_oz\NMCB\O. Data management\1. Preparation\Codebook*

1.10

Data acquisition type	Description	Type	Format	Size estimate	Sensitivity	Terminology standard
[1. Descriptive documents, 2. Reuse data, 3. Measured data, 4. Collected data, 5. Analysis data, 6. Publication or archiving (meta)data]	*[e.g., TMF/ISF or separate Protocol, ICF, agreements, etc.; Data dictionary/codebook, Raw questionnaire data, interview recordings, etc.]*	*[e.g., images, scans, spreadsheet, script]*	*[e.g., .csv, .dicom, .sav, .pdf. odt, .jpg, .sps, .mxf, .rdf, .xml, REFI-QDA, etc.]*	*[e.g., ~1MB, <1GB, xx GB, ~1TB, mention total per type of file format expected]*	*[1. Highly sensitive personal data (interview data/directly identifiable data), 2. Medium sensitive data (encoded subject data), 3. Low/non-sensitive data (standardized questionnaire without specific demographic or medical history data, etc.)]*	*[e.g., ATC for type of medication, LOINC for lab values, SNOMED CT for treatment, ICD-10 for condition, medRA for medical coding, CDISC standards (e.g., SEND standard for pre-/non-clinical data, CDASH for collected clinical data, or SDTM for analysis data, etc.]*
1. Descriptive documents	*- Trial Master File (TMF) / Investigator Site File (ISF) - Protocol, - ICF, - Data processing agreements, etc.;*	*Document*	*.doc .odt, .pdf.*	*< 1 GB*	*3. Low/non-sensitive data*	*NA*
3. Measured data	*- MRI data*	*Images. Scans.*	*.dicom*		*2. Medium*	*See 1.8*
3. Measured data	*- ECG - Heart rate*	*Images. Scans.*	*.7fs*		*2. Medium*	*See 1.8*
3. Measured data	*- Lab data*	*Spreadsheet*	*.xlsx*		*2. Medium*	*See 1.8*
3. Measured data	*- Cognitive functioning data*	*Spreadsheet*	*.csv*		*2. Medium*	*See 1.8*
4. Collected data	*- Blood pressure*	*Spreadsheet*	*.csv*	*<1 GB*	*2. Medium*	*See 1.8*
4. Collected data	*- Weight, - Height - Bioimpedance*	*Spreadsheet*	*.csv*	*<1 GB*	*2. Medium*	*See 1.8*
4. Collected data	*- Handgrip strength*	*Spreadsheet*	*.csv*	*<1 GB*	*2. Medium*	*See 1.8*
4. Collected data	*- Pulse oximetry*	*Spreadsheet*	*.xlsx*	*<1 GB*	*2. Medium*	*See 1.8*
4. Collected data	*- Questionnaire data*	*Spreadsheet*	*.csv*	*<1 GB*	*2. Medium*	*See 1.8*

Reuse of existing data
☐ Not applicable

1.11 *Specify the source that is used to acquire the existing data: …*

1.12 ☒ The reuse of existing data for this study is covered by the subject's informed consent

1.13 ☐ The party that delivers encoded data remains responsible for the subject identification log for their own subjects

Measured data
☐ Not applicable

1.14 *Specify the device that generates the measured data:*

- *MRI*
- *ECG and heart rate by **VU-AMS Core***
- *Blood pressure by **Omron M7 Intelli IT***
- *Weight, height, and bioimpedance by **Tanita MC-580***
- *Handgrip strength by **Jamar Smart Handdynamometer***
- *Pulse oximetry by **Nellcor PM 10n***
- *Cognitive functioning by **Amsterdam Cognition Scan (ASC)***

1.15 ☐ No user training is required; the device that generates the data is self-explanatory

☐ No separate user training is required; users are already acquainted with and trained in using the device for the type of measurements that are required for this study

☒ Users are trained in using the device /performing the lab measurements in the way needed for this study this is documented

A Study Training Log is developed to document the relevant training content and activities.

- File: Paper
- Location: Rosalyn, Amsterdam, the Netherlands.

	Device manuals are stored in: G:\divjk\mp_oz\NMCB\K. Other relevant documents\Device manuals\ - Instruction manual MC580 & printer set up (EN) 2021 (1).pdf - Manual - Handgrip strength - jamar-smart-userguide_012221.pdf - Manual - VU-AMS Core Manual 11-03-2024.pdf - Nellcor PM10N Oximeter manual.pdf - Omron Intelli M7.pdf
Data collection	☐ Not applicable
1.16	☒ An electronic CRF is used to collect all or a part of the data, a copy of the blank CRF pages is kept as back-up ☐ A paper CRF is used to collect all or a part of the data
1.17	*Specify the name of the data collection system:* - *Castor EDC System is utilized for creating eCRF and questionnaires.* - *Amsterdam Cognition Scan (ACS) system is utilized for collecting cognitive functioning data.* ☒ Licensing and a Processing agreement have been arranged
1.18	☐ The developer of the eCRFs and/or questionnaires in the data collection system is already acquainted with the system ☒ The developer of the eCRFs and/or questionnaires in the data collection system is trained in using the system and this is documented. *Castor Academy has been used to get acquainted with Castor. It is recorded in the Study Training Log document (see 1.15)*
1.19	☒ The database was designed before it was built and a data dictionary was created, *specify document name and location of the data dictionary: see 1.9.*
1.20	☒ Validation checks on completeness, correctness and consistency are incorporated in the data collection system and have been documented. Non-WMO*:* – File: F08 Castor eCRF Quality Control checklist - RDM - Amsterdam UMC non-WMO v2.pdf – Location: G:\divjk\mp_oz\NMCB\O. Data management\1. Preparation\Data collection system validation\Quality check\non-WMO WMO*:* – F08 Castor eCRF Quality Control_NMCB MeasQuest.pdf – G:\divjk\mp_oz\NMCB\O. Data management\1. Preparation\Data collection system validation\Quality check\WMO
1.21	☒ The data collection system has been tested by both the study team and an independent party. - *The quality control check will be conducted by RDM, and the form "Castor eCRF Quality Control checklist" will be filled in.* - *The test process will be documented following the "User Acceptance Test" template.* - *Test date, name, and function of testers will be recorded.* ☒ The test findings, follow-up of the findings and final approval are documented, *with all documents stored in:* - G:\divjk\mp_oz\NMCB\O. Data management\1. Preparation\Data collection system validation
1.22	☒ Access to the data collection system is based on individual login with only the necessary access rights ☒ Access to the data collection system is managed (under supervision of) and documented by the PI *To keep track of all authorizations, the authorization log will be developed.*
1.23	☒ The data collection system logs the identity of the persons using the system *Castor EDC system contains such function.* *ACS is filled in by participant using Participant ID*
1.24	☒ Procedures for data collection are included in the data collection system ☒ Procedures for data collection are documented in a manual, *specify document name and location:* - *File - Device Data Transfer SOP Complete v2.pdf* - *Location - G:\divjk\mp_oz\NMCB\C. Protocol and amendments\4. Research Nurse SOP's\Data handling* - *File - Castor Survey Send SOP v2.pdf* - *Location - G:\divjk\mp_oz\NMCB\C. Protocol and amendments\4. Research Nurse SOP's\Other*

1.25	☐ No user training is required; the data collection system is self-explanatory ☐ No user training is required; users are already familiar with the data collection system (only non-WMO) ☒ Users are trained in the data collection system, and this is documented, *specify document name and location: …* *See protocols in 1.18 and 1.24.*
1.26	☐ For each study-specific data collection, source documentation is available ☒ For the following parts of data collection, no source documentation is available, as described in the protocol; *specify:* *Height measured with tape measure, handgrip strength measured with analogue device, pressure pain threshold measured with VAS and verbal communication,*
Data storage (both during study and for archiving)	
1.27	What data (raw files, intermediate files, final files, subject identification log (key file)) is stored on which location during the study?

Location	What data?	
On the department's G-drive (location AMC) See 1.6.	☒ Raw ☐ Final	☒ Interim ☒ Key
On a data storage facility at Amsterdam UMC or at an Amsterdam partner In Castor, and SQL Server	☒ Raw ☐ Final	☒ Interim ☐ Key
On an external data storage facility Not Applicable.	☐ Raw ☐ Final	☐ Interim ☐ Key

1.28	☒ The size of the data during data collection and the size when archiving the project can be determined **only when** the data of the first participant is collected. ☒ Budget is allocated both for storage during the study as well as for archiving upon completion of the study ☒ Budget is allocated for data management activities and creating a FAIR dataset
1.29	☒ The study-specific folder with data files is only accessible by study team members *All data files will be organized by pre-determined file structure within the 'NMCB' folder on the department's network drive (G drive).* ☒ Access to study data is managed and documented by the PI *See 2.1.*
Subject identification log	☐ Not applicable
1.30	☒ The subject identification log(s) is/are kept separate from other study related data.
1.31	☐ In multicentre studies the site-specific subject identification log(s) is/are kept on site only and will not be shared centrally or with other sites.

Legal issues/agreements/matters	☐ Not applicable
1.32	For collaboration with the research partners or for data transfer or data sharing, written agreements on data management, privacy, data ownership and intellectual properties are made, namely: ☒ A collaboration agreement with consortium partners or project members, i.e., consortium agreement or clinical trial agreement, which is stored in G:\divjk\mp_oz\NMCB\P. Privacy ☐ An agreement with other parties, e.g., data sharing agreement or material transfer agreement, *specify: …*
Additional information on Phase 1: Study preparation (provide item number)	

Phase 2: Data collection	
General	
2.1	☒ A site signature and delegation log of all people involved in the data collection is kept by the PI *The document 'Site signature and delegation log' is being filled in:* - See ISSF document.
2.2	☐ *If applicable:* a procedure for deblinding is in place and has been documented
Reuse of existing data / use of measured data ☒ Not applicable	
2.3	☒ The reused existing data and/or the raw, measured data are stored as read-only file and a copy is made for further processing
Externally acquired data ☐ Not applicable	
2.4	☒ Identifiable data is removed or encoded by the external party, prior to sharing the data
2.5	☒ Data are transferred in a secure way by: - SURF FileSender, - Research Data Platform (RDP), - *A customized application being developed to connect Amsterdam Cognitive Scan and Castor so that data transfer can be automated.*
Quality control	
2.6	☒ Checks on completeness, correctness and consistency are built into the system (as specified in the *Data Validation and Derivation Plan)* ☒ Non-automated checks such as manual checks and data monitoring are performed and are documented *See 1.20.*
2.7	☐ Completion of multicentre data collection is signed off by the local PI per patient and the coordinating PI for completion of data collection
Change control ☐ Not applicable	
2.8	How are changes to the **data** handled? ☐ Documented on the paper data collection tool (pCRF or questionnaire) There is no data collected on paper. ☒ Audit trail or 'track changes' functionality in the applied system The Castor system has the audit trail function. In the Access Application, such audit trail function is under development and processed with the technical data steward. ☒ Reason for change is documented, e.g., - 'Confirm changes' setting in Castor is used - Change of Castor system is documented in: G:\divjk\mp_oz\NMCB\O. Data management\1. Preparation\Change log ☐ Other change control; *specify: …*
2.9	How are changes to the **design** of the data collection handled? ☐ By creating a new version of the paper data collection tool (pCRF or questionnaire) ☒ Audit trail or 'track changes' functionality in the system ☐ Other change control procedures; *specify: …* ☒ All changes in the design will be documented to ensure an impact assessment of these changes is performed
Locking a data collection ☐ Not applicable	
2.10	*Describe how the data collection system is locked:* ☒ Using the locking functionality in the system ☐ Other; *specify: …*
2.11	☒ Approval and reason for locking have been documented. *A Lock Study Data document should be developed for the future use:* – *NMCB Lock study data.pdf* – *Location: G:\divjk\mp_oz\NMCB\O.Data management\2. DATA COLLECTION*
2.12	☐ The Statistical Analysis Plan is finalized, prior to (deblinding and) analysing the data Not applicable.
Additional information on Phase 2: Data collection (provide item number)	

Phase 3: Processing & statistical analysis	
Export to the data processing and statistical environment	☐ Not applicable
3.1	☒ The data are stored in a generic and machine actionable format; *specify:* CSV ☒ Data are also stored in another format; *specify:* See 1.10.
3.2	☐ *Specify the software system including version number, applied for processing and statistical analysis: …*
Performing data processing and statistical analysis	☐ Not applicable
3.3	☒ The acquired data are stored as read-only file and a new file is created for further processing and statistical analysis
3.4	☒ All data processing and analysis is programmed in syntax or script files
3.5	☒ Descriptive comments are added to the syntax or script files
3.6	☒ Data sets and syntax or script files are placed under version control
3.7	☒ Data corrections in this phase are made in the original source ☒ Data corrections in this phase are programmed in syntax or script files
Sharing data for processing or statistical analysis	☒ Not applicable
3.8	☐ Data processing and analysis by an external party is covered by the informed consent
3.9	☐ Data are transferred in a secure way by SURF Filesender or *specify which secure transfer is used: …*
Additional information on Phase 3: Processing & statistical analysis (provide item number)	

Phase 4: Writing & publishing	
Organizing files	
4.1	☐ For each manuscript a structured subfolder has been created NA. Research projects should develop their own DMPs to reuse our data for conducting research, so the manuscript subfolder is needed thereafter.
Findability of the data set	
4.2	☐ The manuscript will be published in an Open Access journal that provides a PID (e.g., a DOI or URN) ☒ The information regarding the data collection can be found through a catalogue or repository; *specify*: - *DataverseNL (https://dataverse.nl/) for manuscript or relevant artefacts in a research project.* - *Amsterdam Cohort Hub (ACH) Data Catalogue (https://aumc-catalogue-test.molgeniscloud.org/catalogue/ssr-catalogue)* - *Own data portal to be built on the NMCB website for data request.* ☒ This catalogue or repository creates a PID - *Yes. Both DataverseNL and ACH Data Catalogue creates PIDs.* ☒ This catalogue or repository has a CoreTrustSeal: - *DataverseNL has a CoreTrustSeal* - *ACH Data Catalogue is still under development* ☒ To make my (meta)data findable, we will crosslink any online sources where applicable *(e.g., ORCIDs of researchers, PIDs of related publications or repository references within the project, trial registry numbers, project website, etc.)* ☐ I will not publish my metadata and/or data; *explain why not …*
Additional information on Phase 4: Writing & publishing (provide item number)	

Phase 5: Data sharing and archiving	
5.1	What will be published?

	☒ Metadata (see 5.2) ☒ The raw, pre-processed data within A-UMC ☒ The final data *stored within A-UMC, and can be available upon reasonable request* ☐ Other (e.g. Software); *specify type and location: …*
5.2	What metadata is provided about the **study?** ☒ Study protocol ☒ (Amsterdam UMC) metadata schema ☒ Data Management Plan ☒ Statistical Analysis Plan (if developed in the future) ☐ Other; *specify:*
5.3	☒ Data reuse of the dataset is covered in the informed consent procedure The template of Informed Consent Forms (ICFs) is available (see 1.5) and it provides a clear description of what data the participants give permission to. ☒ Information regarding the subset of the data for people who consented to reuse is available *Yes.* ☒ Procedures for withdrawal of consent for reuse have been defined *The standard procedure for withdrawal of consent for reuse is under development. Such procedure should describe which variables need to be changed from yes to no and which variables need to be filled in with further information.* ☒ A more pseudonymized version of the dataset has been created for reuse in consultation with the DPO *Before issuing data to researchers, the entire dataset should be reviewed by Data Privacy Office to see if any variables need further pseudonymization steps and whether there are certain variables that can't be issued together.* ☒ For verification purposes, all data are stored internally (see 5.5 for data access procedures).
5.4	What metadata is provided about the **data**, including processing and analysis? ☒ Documentation on study procedures ☒ Syntaxes or scripts Other; specify: FIP ☒ Data dictionary ☒ Software ☒ Data validation and derivation plan ☐ Hardware
5.5	☐ The research data will be publicly accessible without any restrictions ☒ Conditions for reuse apply will be based on JRDA, which is almost completed. ☒ An embargo period of six months will commence once the clean and structured NMCB data become available. ☐ At the time of the journal article's publication
5.6	☒ For data reuse, a Data Sharing Agreement or equivalent will be set up JDRA for the six research projects are available. For additional requests, further policy is to be decided, either data sharing agreement or JDRA.
Transfer to an external party	☐ Not applicable
5.7	☒ (Copies of) original data and documentation are kept at Amsterdam UMC ☒ All data transfers have been documented
5.8	☒ Data are transferred in a secure way by **SURF Filesender** (or *specify which secure transfer is used:* ***myDRE***)
Digital archiving	
5.9	☒ *Digital data and documentation will be preserved for 20 years. This includes:* ☒ metadata ☒ raw data files ☒ final data files
5.10	*Specify the location of the digital archive: See 1.6 and 1.27*
5.11	☒ A subject identification log is archived and kept separate from other study related data. This does not conflict with the subject's informed consent
Paper archiving	☐ Not applicable
5.12	☒ *Paper documentation will be preserved for 20 years*
5.13	*Specify the physical location of the paper archive:* See 1.6
Additional information on Phase 5: Data sharing & archiving (provide item number)	